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Evolution and evolvability of rifampicin resistance across the bacterial tree of life

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Predicting the ability of bacteria to develop antibiotic resistance is challenging, especially for the vast majority of species for which no experimental data are available. Here, we investigated the evolvability and intrinsic presence of rifampicin resistance across the bacterial tree of life. We compiled a comprehensive panel of known rifampicin resistance mutations, comprising 57 amino acid substitutions within the gene *rpoB*. We then screened more than 18,000 genomes from all major bacterial groups for the presence of those mutations and determined which mutations can evolve through point mutations. Our results demonstrate that although the evolvability of individual mutations varies considerably across species, overall predicted evolvability is high and relatively homogeneous across bacterial taxa. Rifampicin resistance mutations are present intrinsically in 8% of species that tend to be phylogenetically clustered. Our analysis provides a global picture of the mutational landscape of rifampicin resistance, affording insight into existing observations and informing future discoveries such as the identification of probiotics.

antibiotic resistance | evolvability | mutational spectrum | intrinsic resistance | rifampicin

Significance

How much do bacteria vary in their ability to evolve resistance to antibiotics? Answering this question is difficult because most research focuses on a small minority of species that are either pathogens or model organisms. Rifampicin is a widely used antibiotic for which extensive data on resistance mutations exist. Comparing this data against the genomes of over 18,000 species of bacteria, we find i) that around 8% of species naturally harbor rifampicin resistance mutations and ii) that although the spectrum of mutations available differs substantially across the bacterial tree of life, the total number available to each species is remarkably similar. Our predictions broaden our fundamental understanding of the distribution of antibiotic resistance across bacteria.

The evolvability of a biological system refers to its ability to respond to natural selection (1, 2). Understanding evolvability is essential for predicting adaptive responses to environmental challenges (3, 4). It is hence relevant to a wide range of fundamental and applied problems in biology, but perhaps nowhere more so than in tackling the evolution of antibiotic resistance (4, 5).

Curtailing antibiotic resistance is predicated on our ability to predict the evolutionary trajectory a microbial lineage is most likely to take (6). A critical first step toward this goal is to determine the spectrum of resistance mutations available to a given lineage and the number of independent pathways it has to access them. With the exception of the most well studied clinical pathogens and laboratory strains, whose resistance profiles have been extensively screened (e.g., refs. 7–11), this information is not readily available for the vast majority of bacteria. Nevertheless, the genomes of known resistant lineages and mutants might still be leveraged to predict the resistance evolvability of understudied taxa.

The mapping of known resistance mutations to unscreened bacteria may also serve to identify taxa that are already intrinsically resistant to certain antibiotics. This is to say we may be able to use current knowledge to make predictions on both the potential to evolve resistance and the preexistence of resistance mutations. Given that most antibiotics are derived from compounds produced naturally by bacteria and other microbes (12), there are likely to be large numbers of species with unrecorded intrinsic resistance to various antibiotics. Identifying taxa that are intrinsically resistant holds promise not just for our fundamental understanding of bacterial diversity but also potentially in the design of probiotics to accompany antibiotic treatment (13).

In the quest to translate resistance from known to unknown taxa, it is reasonable to expect that the robustness of predictions across taxa will be greatest for antibiotics that 1) target highly conserved sites; 2) cannot be evaded by too large a variety of resistance mechanisms; and 3) whose resistance evolution has already been studied in some depth. One drug that ranks highly for these three criteria is rifampicin (RIF), a broad-spectrum antibiotic that is used to treat a range of bacterial infections and is a first-line drug for the treatment of tuberculosis (caused by *Mycobacterium tuberculosis*, Mtb) (14, 15). The drug targets the beta subunit of the RNA polymerase, binding close to its active site and blocking elongation of nascent RNA transcripts, thereby leading to premature termination of transcription and ultimately cell death. Resistance to RIF arises predominantly through single point mutations in *rpoB*, the gene coding for the beta subunit of the RNA polymerase (15, 16). Prior work in several species has established that almost all RIF resistance mutations arise in one of four, relatively small clusters

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within *rpoB*: I, II, III, and N. Of these, cluster I is the largest and contains most reported resistance mutations, including in clinical isolates of RIF resistant Mtb (17).

We set out to predict the evolvability of RIF resistance across the bacterial tree of life. Specifically, using a comprehensive inventory of reported resistance mutations from the literature, we screen the genomes of >18,000 bacterial species spanning all major phyla for the evolvability and preexistence of RIF resistance. We find that although total evolvability (number of resistance mutations accessible though a single nucleotide change) is largely invariant across species, the spectrum of mutations available varies considerably. In addition, we find that the predicted distribution of existing RIF resistance is clustered phylogenetically, with several clades uniformly possessing known RIF resistance mutations. This includes several groups for which there is direct, but until now unexplained, evidence of RIF resistance.

Results

Mutations Reported to Confer RIF Resistance. We compiled data from 115 studies reporting a total of 1,366 RIF resistance mutations within the *rpoB* gene (Datasets S1 and S2). Merging homologous residues reported in different studies and/or species produced a list of 358 distinct single amino acid substitutions at 146 positions. These mutations were reported across 75 bacterial species and included both mutations obtained in experimental studies (mostly mutant screens) and in clinical or environmental isolates (see Fig. 1 for an overview). Mutations 526y and 526R (all positions in *Escherichia coli* coordinates) had the highest frequencies (reported in 30 and 26 species, respectively), and position 526 is also the position with the greatest number of reported mutations (Fig. 1*A*). The largest number of mutations were reported in Mtb, *Salmonella enterica*, *E. coli* and *Streptococcus pneumoniae* (Fig. 1*B*).

From this large pool of reported mutations, we sought to construct a panel of mutations that were likely to robustly confer resistance across species. We therefore discarded mutations that were reported in fewer than three independent studies or three species. After applying these filters, 57 mutations at 19 *rpoB* amino acid positions were included in our panel. Most of the mutations in our panel were reported in both mutant screens and clinical or environmental isolates. Mutation 531 N is the only mutation in our panel that was reported in natural isolates only but not in mutant screens (*Discussion*).

These 19 amino acid residues are all in close proximity to the binding pocket of RIF within the 3D structure of wildtype *E. coli* K12 RpoB (see file *rpoB_structure.html* in our repository at [https://github.com/JanEngelstaedter/RIFR_evolvability/releases/tag/publication_release]). Moreover, these amino acid positions are generally highly conserved (*SI Appendix*, Fig. S1). Together, these observations support the notion that the mutations in our panel confer RIF resistance in the majority of bacterial species.

Comparing our panel to mutations reported in Mtb—the bacterium with most reported resistance mutations and for which RIF is clinically most relevant—we note that 36 of the mutations in our panel have previously been reported in this species. We also cross-validated the mutations in our panel using an independent dataset of mutations that were reported (18) to be associated with elevated minimum inhibitory concentrations (MICs) of RIF, based on an in vitro screening of more than 15,000 clinical isolates of Mtb (19). Most (25 out of 34) of the mutations associated with elevated MICs in Mtb are also included in our panel (*SI Appendix*, Fig. S2). Of the remaining nine mutations, all except one were considered "borderline variants" by the authors of ref. 18 because they were only associated with weakly increased MICs. Of note, we would still recover the same 25 mutations associated with elevated MICs in Mtb even if we had excluded all Mtb data in the construction of our mutation panel. These findings again corroborate our mutation panel's prospects for cross-species predictions.

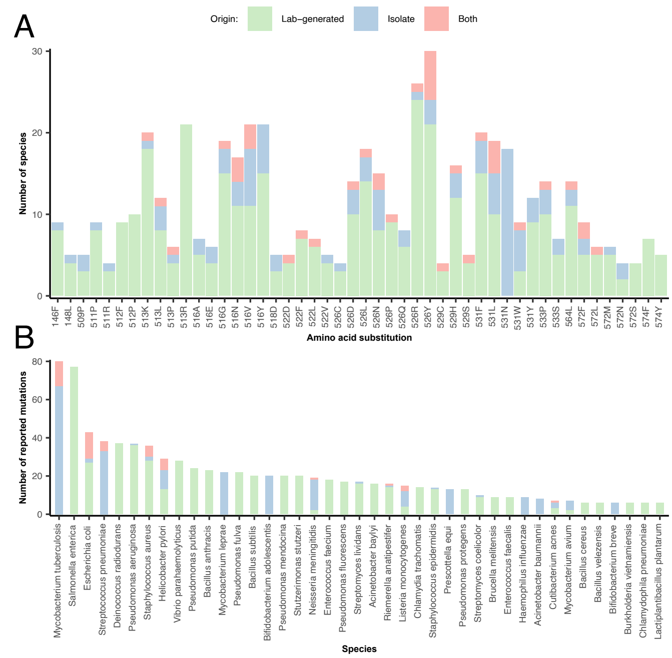

Fig. 1. RIF resistance mutations within *rpoB* reported within the literature. (*A*) The number of species in which different resistance mutations were reported, colored by whether the mutations were generated in experimental studies (mostly mutant screens), in clinical/environmental isolates, or in both settings. Only mutations reported in at least four species are shown. (*B*) The number of mutations reported across species, with the same color coding as in (*A*). Only species with at least six reported mutations are shown.

The Phylogenetic Distribution of Predicted Intrinsic and Evolvable RIF Resistance Across Bacteria. We next extracted *rpoB* sequences from 18,469 annotated reference or representative bacterial genomes obtained from the NCBI genome database (20). After applying quality-control filters to these sequences, we retained 18,238 high-quality *rpoB* sequences from 18,127 species (*SI Appendix*, Fig. S3). (91 species carried more than one *rpoB* copy; see below.) We then screened each sequence for both the presence and evolvability of the 57 mutations in our panel of reported RIF resistance mutations.

Based on this screen, we predict 1,438 species (8% of the species in our dataset) to be intrinsically resistant to RIF because they carry at least one of the resistance mutations from our panel (Dataset S3). These species are spread across the entire bacterial phylogeny but cluster (permutation test: $P < 0.001$) in distinct bacterial clades (Spirochaetia, Bifidobacterales, Mollicutes, Erysipelotrichales, and others) (Fig. 2). Conversely, some large clades (e.g., classes Cytophagia, Sphingobacteriia, and Chitinophagia within the phylum Bacteroidea as well as the Enterobacterales) appear to be devoid or almost devoid of any intrinsically resistant species.

Evolvability, here defined as the total number of RIF resistance mutations (at the amino acid level) that can arise through a single point mutation (at the nucleotide level), varies between 25 and 47

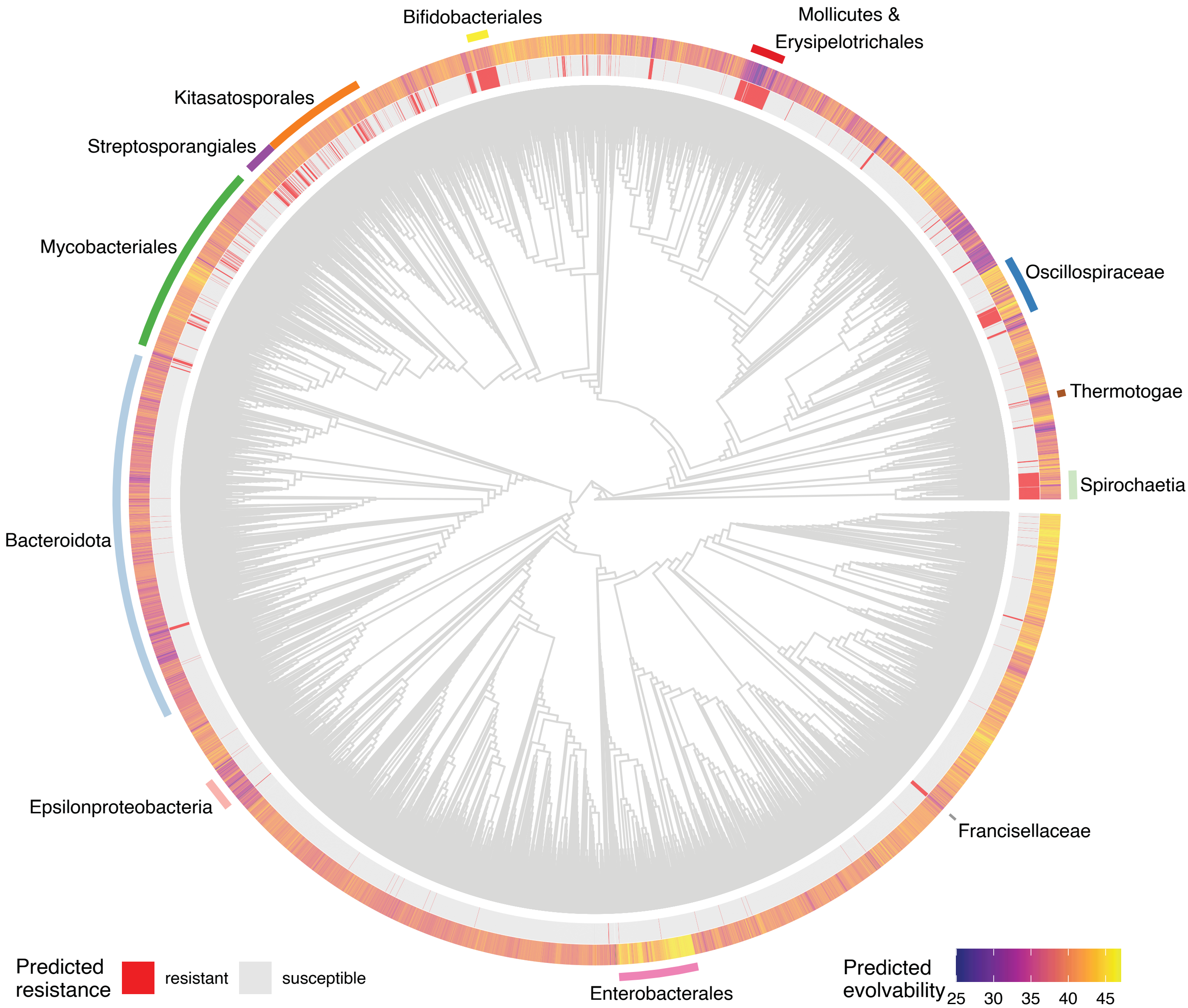

Fig. 2. The phylogenetic distribution of predicted intrinsic resistance and evolvability of RIF resistance. The phylogenetic tree is a subtree of the phylogenetic tree available on the Genome Taxonomy Database (21), containing the majority (15,521) of our screened species. The inner circle shows predicted intrinsic resistance as red lines (species with at least one amino acid residue from our panel of reported resistance mutations). The outer circle shows the predicted evolvability of RIF resistance (number of amino acid substitutions from our panel that are accessible via a single point mutation).

across the bacterial tree of life (95% interquantile range: 33 to 45). The upper limit of this observed range in evolvability is close to the theoretical maximum of 50 (this theoretical maximum, which is considerably below the total number of 57 mutations, arises because many reported mutations are at the same site and are impossible to all evolve from any one codon through a single nucleotide substitution). As observed for intrinsic resistance, there is a clear phylogenetic signal in the distribution of evolvability, with some clades displaying distinctly high or low values (Pagel's $\lambda = 0.964$; $P < 0.001$ and Blomberg's $K = 0.211$; $P = 0.001$).

Fig. 3*A* shows the distribution of evolvability and intrinsic resistance for the largest bacterial classes. Almost all species within the Spirochaetia carry the 531N mutation and are hence predicted to be intrinsically resistant to RIF. Similarly, almost all members of the Mollicutes are predicted to be resistant, but in this case due to the presence of 526 N (with the exception of 25 species harboring 526Q). This class also exhibits very low evolvability, indicating that Mollicute species generally cannot easily acquire many of the other known resistance mutations. Other classes comprising many species predicted to be intrinsically RIF resistant include the Actinomycetes (638 species), and the Clostridia (144 species). Aside from well-known genera within the Spirochaetia (e.g., *Borrelia* and *Treponema*) and Mollicutes (e.g., *Spiroplasma* and *Mycoplasma*), unrelated genera such as *Streptomyces* are also predicted to comprise many or even mostly resistant species (Fig. 3*B*). Notably, intrinsic resistance within *Streptomyces* appears to have arisen independently many times as not all members of this genus have RIF resistance mutations despite having a close phylogenetic relationship (*SI Appendix*, Fig. S4; see also *Discussion*). Another, small class that is predicted to be almost universally resistant is the Erysipelotrichia (52 out of 54 species; see also *Discussion*). A total of 94 species carry two or more resistance mutations, a pattern particularly common in the genera *Nocardia* and *Nonomuraea*.

There were 90 species with two copies of the *rpoB* gene in our dataset (most prominently belonging to genera *Nocardia*,

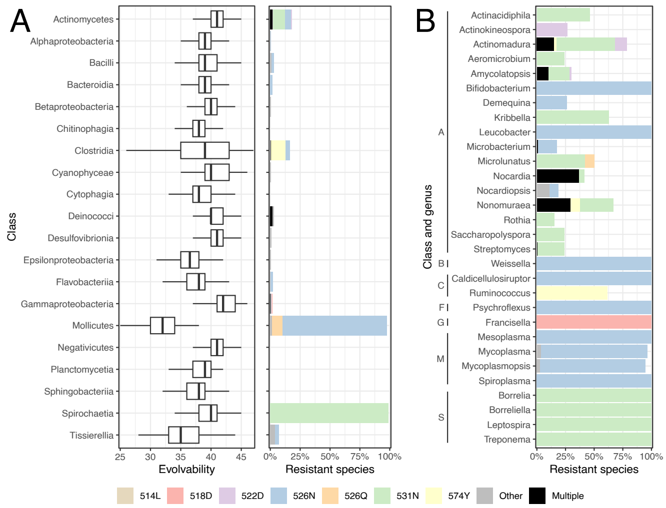

Fig. 3. Predicted evolvability and intrinsic resistance at different taxonomic levels. The *Left* plot in panel (*A*) shows the distribution of evolvability across the twenty most specious bacterial classes. Evolvability is defined as the total number RIF resistance mutations that can arise through a single point mutation. The *Right*-hand plot in panel (*A*) shows the percentage of species predicted to be intrinsically resistant in the same classes, with colors indicating which resistance mutation they carry. Only the six most common mutations are shown explicitly, all other mutations are shown in gray. Panel (*B*) shows the percentage of resistant species across the thirty genera with the highest percentage of resistant species. Only genera with at least twelve species were included. Genera are grouped by class, with the following abbreviations: A = Actinomycetes, B = Bacilli, C = Clostridia, F = Flavobacteriia, G = Gammaproteobacteria, M = Mollicutes, and S = Spirochaetia.

Nonomuraea and *Actinomadura*; one species, *Oceanobacillus jeddahense*, had three copies). Sequence alignment revealed that in some species the two sequences are identical at the main RIF resistance-determining region of *rpoB* (*SI Appendix*, Fig. S5), indicating relatively recent duplication events (or potentially erroneous genome assemblies). However, most pairs of *rpoB* copies exhibit considerable divergence, and in most cases one copy carries one or several resistance mutations whereas the other one does not (*SI Appendix*, Fig. S5).

The Mutational Spectrum of Intrinsic and Evolvable RIF Resistance. Although overall evolvability of RIF resistance varies relatively little across bacterial groups, there is considerable variation when individual mutations are considered (Fig. 4). Some mutations can readily arise in essentially all bacteria, either through one, two, or even three different point mutations. Other mutations can arise in only some but not other species, and some mutations can only arise through a single point mutation in very few species. This variation also persists when mutations are pooled into amino acid positions, with some positions (e.g., 513 and 516) exhibiting consistently high evolvability whereas at other positions (e.g., 522 and 574) no mutation is evolvable in all or even the majority of species. Fig. 4 also highlights which resistance mutations are already present in some species, the most common being 531N (565 species), 526N (546 species), and 574Y (183 species). Stratifying this mutational spectrum taxonomically demonstrates that heterogeneity in evolvability can differ considerably across classes (*SI Appendix*, Fig. S6). For example, mutation 509R can evolve via three different point mutations in most members of the Deinococci but this same mutation cannot evolve through a single point mutation in many other classes.

We can shed light on this variation in evolvability by considering mutational codon networks, i.e. networks at a specific site where only codons that differ by a single nucleotide (and thus are evolvable from each other) are connected. Fig. 5 shows an example for position 534. At this position, a single amino acid residue (aspartic acid) has been reported to confer RIF resistance (534 d). All species in our dataset carry a glycine (G) residue at this position, but all four codons coding for glycine are used across species (at frequencies of 26%, 40%, 17%, and 17% for GGA, GGC, GGG, and GGT, respectively). Only the GGC and the GGT codons can mutate toward codons coding for aspartic acid in a single step, which explains why many species (43%) cannot evolve resistance at position 534. A more complex scenario, at position 531, is shown in *SI Appendix*, Fig. S7. At this position, almost all species (96%) carry a serine residue. How many and which of the six reported resistance mutations (531C, F, L, N, W, and Y) can evolve in a given species depends on which of the

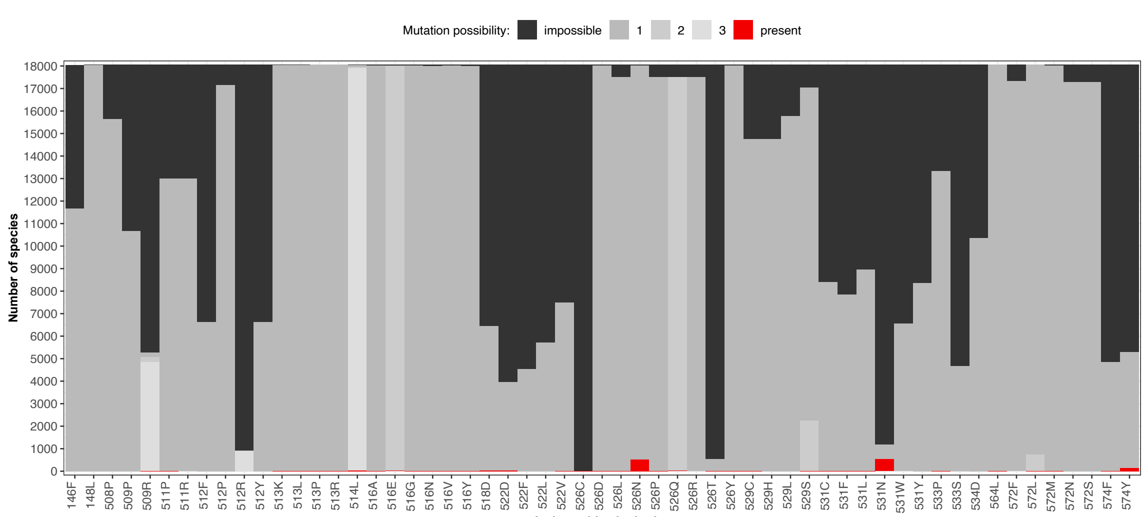

Fig. 4. The mutational spectrum of intrinsic and evolvable RIF resistance. For each of the 57 amino acid residues in our panel of screened RIF resistance mutations, the plot shows the number of species in which this mutation is either present (red), evolvable through a single point mutation (gray), or impossible to evolve through a single point mutation (black). Evolvability is further distinguished by different shades of gray according to whether the amino acid mutation can arise through one, two, or three different point mutations at the respective codon. Species with multiple *rpoB* sequences were excluded from this plot.

six codons coding for serine is employed. For example, 531W can only evolve in species with a TCG codon, whereas 531F and 531Y can only evolve in species with a TCC or TCT codon.

Cross-Validation with a High-Throughput Screen in *E. coli*. Finally, we sought to complement the results derived from our panel of reported mutations with data from a recent high-throughput screen by Yang et al. (11). In this study, the authors employed multiplex automated genome engineering (22) to create all possible single amino acid residues at the RIF binding site of *rpoB*, and through bulk competition estimated conferred RIF resistance for each mutation. As expected, there is substantial overlap between our panel of reported mutations and the mutations in Yang et al. (*SI Appendix*, Fig. S8). However, there are also many mutations not in our panel that Yang et al. predict to confer resistance. Conversely, there are also mutations in our panel that do not show up as resistant in Yang et al. Some of the mutations in the latter group (at positions 146, 148, 508, 509, and 574) were outside of the region screened in Yang et al. (11). We suspect that other mutations in our panel may impose strong fitness costs in *E. coli* and may for this reason not have spread in the experiments reported by Yang et al.

SI Appendix, Fig. S9 shows that the resistance mutations predicted in Yang et al. (11) that are absent from our panel of reported mutations are generally characterized by low evolvability. With few exceptions, most of these amino acid residues cannot arise via a single point mutation, which explains why they have not been reported in previous mutant screens. Moreover, with one exception (525R), these mutations are either completely absent or very rare (<0.2% frequency) within the species we screened. These results indicate that although there are likely many more resistance mutations theoretically possible than previously reported in mutant screens and natural isolates, these mutations are neither common in nature nor can they arise easily from existing genetic variation.

Discussion

Predicting antibiotic resistance, a key challenge in medicine, public health, and agriculture, requires knowledge of both selective pressures (driven mostly by antibiotic use) and evolvability (the amount of genetic variation for resistance, or the ease by which such variation can be generated). Focusing on the latter, we have analyzed *rpoB* sequences from more than 18,000 species, enabling us to identify, explain, and predict patterns of past and future RIF resistance evolution across the bacterial tree of life. Our approach underscores both the power and the limitations of leveraging genetic information across taxa.

Our results indicate that overall evolvability of RIF resistance is relatively homogeneous across the bacterial tree of life. Through a single nucleotide substitution, most species have immediate evolutionary access to between 33 and 45 resistance mutations, out of the 57 mutations that we screened. This means that the genetic code imposes only minor constraints on RIF resistance evolution (see also ref. 23), and any large differences in mutation rates toward RIF resistance across species are unlikely to be explained by variation in evolvability as defined here. However, despite the homogeneity in total evolvability, there is considerable variation at the level of many individual mutations.

This variation can be important because individual RIF resistance mutations, even at the same position, can have very different phenotypic effects. Examples include different degrees of reduced competitive ability ("fitness costs") in *Riemerella anatipestifer* (24), a wide range of minimum inhibitory concentrations and

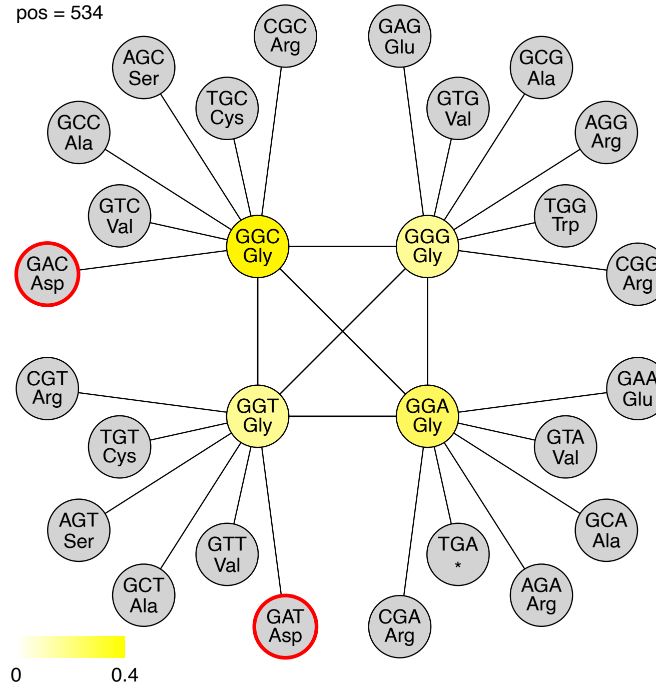

Fig. 5. Incomplete codon network at *rpoB* amino acid position 534. Circles represent codons and lines connect codons that are separated by a single nucleotide difference. The yellow color scale indicates the fraction of species that have a particular codon, with gray indicating that no species has the respective codon. Red circles indicate amino acids residues that have been reported to confer RIF resistance. For clarity, edges between unoccupied (gray) codons are not shown.

pleiotropic heat-stress resistance induced by different mutations at position 572 in *E. coli* (25), and divergence in patterns of carbon usage in *Bacillus subtilis* mutants carrying different *rpoB* mutations at positions 513 and 526 (26). Moreover, different RIF resistance mutations can also differ in their epistatic interactions with other resistance mutations (27, 28). For instance, while both positive and negative epistasis in competitive ability has been reported between RIF resistance mutation 526L in *rpoB* and various streptomycin resistance mutations in *rpsL*, no such epistasis was reported for 526N (27). Finally, different resistance mutations at the same position may also determine future evolvability in different ways, for example by determining possible trajectories toward increased resistance (28) or reduced fitness costs [i.e. compensatory evolution (29)]. Qi et al. reported very efficient compensatory evolution for the costly H526R mutation in *Pseudomonas aeruginosa*, but only incomplete compensatory evolution for H526Y (30).

Around 8% of the species we screened carry at least one RIF resistance mutation. On this basis, we predict that several high-level taxa (e.g., Spirochaetia, Bifidobacterales, Mollicutes, and Erysipelotrichales) as well as many smaller taxa are comprised entirely or predominantly of species that are naturally RIF resistant. RIF resistance mutations are likely to have occurred in the ancestors of many of these lineages in response to chemical compounds in their habitat that are structurally related to RIF. Similar to other drug-like metabolites, such chemicals are naturally released by bacteria to harm competitors, selecting for resistance in both producers and their targets (31–33). RIF is a semisynthetic antibiotic that belongs to the rifamycin class of antibiotics. It is a derivative of rifamycin B, which was discovered in *Amycolatopsis rifamycinica* (then classified as *Streptomyces* mediterranei) (15). Not surprisingly then, *A. rifamycinica* carries two RIF resistance mutations in *rpoB* (522D and 531N), and in fact around one third of the species in this genus carry one or both of these mutations. Related compounds belonging to the larger group of ansamycins that also exhibit antibacterial activities by inhibition of RNA polymerases are produced by other soil bacteria such as *Streptomyces* spp. and *Micromonospora halophytica*, as well as some marine and plant-associated bacteria (34). While it may be tempting to assume that all species carrying RIF resistance mutations evolved them in response to the presence of RIF-like compounds in their environment, these mutations could also have evolved in response to other selective pressures such as temperature (25), or neutrally.

It is important to stress that the presence of a known, RIF resistance conferring mutation in a species does not inevitably result in phenotypic resistance. For example, 526N is ubiquitous in the Erysipelotrichales and Bifidobacterales, thus predicting natural RIF resistance in these groups. In accordance with this prediction, several studies have reported an increase in the frequency of members of the Erysipelotrichales in gut microbiota under RIF treatment (35, 36), but a corresponding increase in frequency has not been observed for species of *Bifidobacterium* (35). Similarly, *Francisella tularensis* appears to be susceptible to RIF despite harboring resistance mutation 518D (37). These examples indicate that at least some of the known RIF resistance mutations can become conditionally inactive through epistatic interactions, either with other sites within *rpoB* or with other genes. Investigating the extent and nature of these epistatic interactions, including careful experimental validation, will be an important task for the future.

One of the most notable *rpoB* mutations on which our investigation sheds light is 531N (asparagine). This mutation has previously been reported to be responsible for intrinsic resistance in several *Streptomyces* (38, 39) and some spirochaete species (40–42). Our analysis shows that this mutation is the most frequent intrinsic resistance mutation, being present in 561 species (3.1% of all species in our dataset). In particular, 531N is universally present in the Spirochaetia and in a diverse range of species within the Actinomycetia. Surprisingly though, 531N is the only mutation in our panel that has not been reported in a mutant screen (only in natural isolates). We can explain this seemingly paradoxical result by our finding that evolvability of 531N is exceptionally low (Fig. 4). The majority of species (>93%) carry a TCA, TCC, TCT, or TCG codon at position 531, all coding for serine. These codons are more than one mutational step away from the two codons coding for asparagine (AAC or AAT). How then has 531N evolved in so many species? We speculate that there could be two major pathways enabling species to evolve the 531N residue. First, some species (≈3%) have codon AGC or AGT (also coding for serine) at this position, from which an asparagine-coding codon can evolve in a single mutational step. Second, a species could first evolve a tyrosine residue (531Y, codon TAT or TAC) and from there, in a second mutational step, evolve 531N. 531Y is a resistance mutation reported in several mutant screens (10, 26, 38, 43–47) but in our screen is not predicted to be intrinsically present in any species. It appears then that 531N is a mutation that is hard to evolve for most species but can confer strong resistance without imposing strong pleiotropic fitness costs, relative to easier-to-evolve mutations.

Several caveats and limitations in our approach merit consideration. First, our panel of RIF resistance mutations involves an inherent trade-off between comprehensiveness (including as many reported mutations as possible) and robustness (ensuring that no erroneous or very species-specific mutations are included). Our cross-validation with an independent dataset (11) indicates that our filtering options navigated this trade-off reasonably well. Second, we only considered point mutations in *rpoB*. Both insertion and deletion mutations in *rpoB* have been reported to confer resistance to RIF as well (7, 10). However, these mutations tend to be rarer and less repeatable than point mutations, so that assessing their evolvability would be difficult. Resistance to RIF can also be caused by other mechanisms, including various enzymes that deactivate RIF, efflux pumps (reviewed in ref. 16), and HelR proteins that dissociate RIF from the RNA polymerase (reviewed in ref. 48). Third, we have employed deliberately simple, binary measures of evolvability in our study, quantifying the number of mutations that can arise through a single nucleotide change and the number of nucleotide changes producing a mutation. A more fine-grained measure would also include differences in mutation rates at the nucleotide level, in particular accounting for higher transition than transversion mutations as well as higher mutation rates at specific sites (49, 50). However, including such differences would still not capture differences in baseline mutation rates across species, which are largely unknown. Finally, we have (with very few exceptions) only considered a single genome per species. This means that there may be species not predicted to be resistant to RIF that are in fact polymorphic for RIF resistance. This is trivially true for many pathogens that are exposed to RIF but may also be true for nonpathogenic bacteria. In fact, our finding that intrinsic RIF resistance in the genus *Streptomyces* exhibits little phylogenetic clustering (*SI Appendix*, Fig. S4) could be explained by widespread but polymorphic resistance in this group, where type strains with deposited genomes sometimes do and sometimes do not exhibit RIF resistance.

In conclusion, our study provides a phylogenetically broad characterization of the mutational landscape of resistance to RIF, an antibiotic of major clinical significance. The approach and bioinformatics pipeline we developed can readily be extended to other antibiotics where resistance is acquired through point mutations in highly conserved genes, including aminoglycosides and quinolones. Pending experimental validation, our predictions on intrinsic resistance and evolvability of RIF resistance could help assess the impact of RIF on commensal gut bacteria as well as assist in the identification of probiotics as an alternative therapeutic strategy to eliminate antibiotic resistant pathogens.

Methods

Selection of Mutation Panel. We sought to assemble a panel of mutations within the gene encoding the beta subunit of RNA polymerase, *rpoB*, that have been reported to confer resistance to RIF. To this end, we compiled studies which reported single amino acid substitutions resulting in phenotypic resistance to RIF, obtained either experimentally (mutant screens or genetic constructs), or in clinical or environmental isolates. We used Google Scholar to find relevant studies using the following key words: "rifampicin resistance," "RNA polymerase subunit beta," "point mutation," and "amino acid substitution." The following information was then extracted from the obtained studies: species name, origin (experimental studies vs. natural or clinical isolates), as well as mutated amino acids and their corresponding positions. Standardized amino acid positions in *E. coli* coordinates were obtained by alignment of the respective *rpoB* sequence to the *rpoB* reference sequence in (*E. coli* MG1655). To avoid the inadvertent inclusion of incidental mutations which may not robustly reflect the evolution of RIF resistance, we only considered mutations reported in at least three studies and within at least three different species.

Prediction of Presence and Evolvability of Rifampicin Resistance Conferring Mutations. We next determined the existence and evolvability of mutations in our panel across all bacterial species with published genomes. First, we downloaded reference and/or representative genomes at all assembly levels (contig, scaffold, and complete) from the NCBI Genome database (51). The *rpoB* sequence of each genome was extracted from their annotated genomic coding sequences using the following keywords: "rpoB," "DNA-directed RNA polymerase subunit beta," "DNA-directed RNA polymerase subunit beta chain," "RNA polymerase, beta subunit," and "DNA-directed RNA polymerase subunit beta." Only *rpoB* sequences with a length of at least 3,000 base pairs were retained.

To further ensure the robustness of these *rpoB* sequences, we translated and aligned them to the *E. coli* MG1655 reference *rpoB* sequence, extracted the amino acid sequence from positions 501 to 600 (the "core" region encompassing the vast majority of RIF resistance mutations). We calculated the Levenshtein distance (52) between the core sequence region of the focal species to the corresponding homologous region of the *E. coli* MG1655 reference sequence. Sequences with a distance greater than 35 were discarded. In total, 444 out of 18682 *rpoB* sequences were discarded due to the length or Levenshtein distance filter.

The filtered sequences were then screened for the presence and evolvability of mutations in our panel. Specifically, for each combination of a reported resistance-conferring amino acid residue and an *rpoB* sequence, we determined: 1) whether the mutation is already present; and 2) in how many different ways (including zero) the mutation could evolve through a one-step single nucleotide change given the codon at the respective position with the *rpoB* sequence. We summarized these data at the level of *rpoB* sequences to determine overall predicted resistance (at least one resistance mutation present) and overall evolvability (total number of mutations that can evolve in a single mutational step).

Phylogenetic Analyses. In order to investigate the phylogenetic distribution of naturally RIF resistant species as well as their evolvability, we downloaded a large phylogenetic tree of all bacterial groups from the Genome Taxonomy Database (GTDB; https://gtdb.ecogenomic.org/) (21). The GTDB tree is based on genome trees constructed using an aligned set of 120 single bacterial protein markers as well as ribosomal proteins and ribosomal RNA genes. The genomes were reference sequences submitted in NCBI (starting with release 1976) (51) and were independently quality controlled using CheckM (53). We subset the GTDB tree for species included in our final screen. We investigated phylogenetic clustering of resistance and evolvability by calculating Blomberg's K and Pagel's λ statistics (54, 55).

RpoB Structure and Conservation. We obtained the predicted 3D structure of the *E. coli* K12 RNA polymerase and RIF complex from the Protein Data Bank (56, 57), and produced an interactive visualization of the RpoB protein and the RIF molecule. In this visualization, we labeled the 19 amino acid residues with reported mutations in our panel. In order to also include information about how evolutionarily conserved these residues are, we calculated the mean Grantham distance (58) between each amino acid in *E. coli* to the corresponding amino acid across all our screened species. Finally, to obtain a measure of amino acid diversity along the *rpoB* gene, we also calculated the mean Grantham distance across 100,000 randomly chosen pairs of *rpoB* sequences from our screened species.

Cross-Validation Using an Alternative Panel of Experimentally Predicted Rifampicin Resistance in *E. coli*. We corroborated our predictions with a secondary mutational screen using a mutation panel employed by Yang et al. (11). The authors of this article constructed a comprehensive set of *E. coli* mutants (760 mutants) carrying all possible single amino acid substitutions within *rpoB* at positions 510 to 537 and 563 to 572 to investigate the effects of *rpoB* mutations on the efficacy of RIF binding. From these 760 mutations, we selected those that increased in frequency in RIF-containing medium (minimum log2(fold-change) of zero), exempting those mutations that were already included in our own mutation panel. We then screened all *rpoB* sequences for the presence and evolvability of these mutations, in the same way that we did for the mutations in our panel of reported mutations.

Implementation. All analyses were conducted in R version 4.4.1 (59). Bioinformatics analyses were performed using packages rentrez v1.2.3 (60), taxonomizr v0.10.6 (61), Biostrings v2.58.0 (62), pwalign v1.0.0 (63), stringdist v0.9.12 (64), Bio3D (65), MSA2dist (66), and future.apply v1.11.2 (67). Packages ape v5.8.0 (68), castor v1.8.0 (69), ggtree v4.4.0 (70), tidytree v0.4.6 (71), and treeio v1.29.0 (72) were used for phylogenetic analysis and tree visualization. Data wrangling and visualization were performed using packages tidyverse v2.0.0 (73), ggnewscale v0.4.10 (74), GGally v2.2.1 (75), ggpubr v0.6.0 (76), ggh4x v0.2.8 (77), RColorBrewer v1.1-3 (78), patchwork v1.2.0 (79), NGLvieweR (80), and htmlwidgets (81).

Data, Materials, and Software Availability. Code have been deposited in GitHub (https://github.com/JanEngelstaedter/RIFR_evolvability/releases/tag/publication_release) (82). All other data are included in the manuscript and/or supporting information.

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